

Pdf Page Duplicate Identification

Analysis

For Example:

If take 2000 page

2000 page → 2000 image

1) Hash matched:

Layout (structure FingerPrint)

Layout:

Paragraph block

Table position

Heading location

Alignment

So 2000 page only **120 page** similar layout

2) Clip Ai(120 page)

Visual similar

Here layout same but content different

visual :

Font density

Line spacing

Graphics

120 page 20 page visually same

So visually same like resume 1 and resume 2 same but content diff

3) Ocr(20 page)

Here extract the image to text then meaning resume 1 and resume 2 is same but name and address phone different so no duplicate

Flow :

- 1) Upload pdf
- 2) pageLoad
- 3) image convert
- 4) hash → layout fingerprint
Similar layout page list
- 5) clip → only sortlist page
Visual embedding store faiss index
- 6) visual similar page get → faiss search
- 7) if text → raw text
If image → ocr → raw text
- 8) text ai
- 9) Final output
If page 20 and 40 similar means
 - 1) Hash similarity ≈ 0.95
 - 2) Visual Similarity ≈ 0.91
 - 3) Text Similarity ≈ 0.60
- 10) Combine
if final_score > 0.90

Duplicate

For example in this image how to identify as per 3 fingerprint

2) CLIP (Visual AI)

It sees:

Certificate layout

Circle stamp

Temple icon

Kannada header

Line alignment

3) OCR + Text AI

Even with damage:

Some text still same:

Government name

Certificate label

Printed lines